

AI-POWERED CAD ANALYSIS REPORT

Generated: 12 Aug 2026

Geometry to Process to Cost · Fully AI-reasoned

53

BUMPER (automotive fascia)

1691 × 647 × 528 mm · 2059.9 cm³ · AI 5.562 kg / Steel 16.170 kg

Manufacturability: 53/100 · Confidence: Low

§1 — Geometry & Part Summary

Bounding Box	1690.9 × 647.4 × 528.1 mm	Surface Area	16261.7 cm²
Volume	2059.90 cm³	Weight (AI)	5.562 kg
Weight (Steel)	16.170 kg	Weight (Plastic)	2.163 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Free-form surface	492	High	Bspline surfaces forming the styled outer/inner bumper skin
Undercut	4	High	Side-wall undercuts requiring side-actions or lifters to release from a single-direction draw
Zero-draft face	2	Medium	Faces with no draft angle relative to draw direction [0,0,1] — will drag/scuff on ejection
Threaded feature	1	Medium	Likely boss/insert thread locations for fascia mounting clips/brackets
Thin uniform-ish shell wall	1	High	2.53–3.54mm nominal wall typical of PP bumper fascia moulding

§3 — Material Analysis

AI-suggested

Primary Material:

Polypropylene (PP, likely TPO/EPDM-modified impact grade) (78% confidence)

Part name 'BUMPER', huge thin-shell panel (2.5mm nominal wall), free-form styled surface, and fill ratio consistent with a moulded plastic fascia. Automotive bumper fascias are almost universally unfilled/impact-modified PP (TPO) rather than glass-filled PP, because glass fill reduces impact elongation and causes fibre read-through under paint. Resolved from part name/geometry per 'Moulded plastic' guidance.

Alternatives:

PP 30% glass-filled (10%) · ABS (7%) · PC/ABS blend (5%)

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Injection Moulding	injection_moulding	85%	0.0128	Stage-1 classifier and geometry (thin 2.5mm shell, large free-form panel, single hollow shell, 200k/yr volume) strongly favour high-volume thermoplastic injection moulding of a single-cavity bumper fascia tool.
Blow Moulding	blow_moulding	12%	0.0200	Fill ratio 0.0036 (enclosed hollow shell) superficially resembles a blow-moulded hollow part, but the styled free-form exterior with local wall variation (2.53–3.54mm) and typical bumper-fascia design is far more consistent with injection moulding of an open shell panel than a sealed blow-moulded vessel.
Thermoforming	thermoforming	3%	0.0150	Possible for a simplified cover panel but unlikely given the surface detail and undercut features implied by tooling.

§5 — Manufacturability Risks (Score: 53/100)

Severity	Feature / Area	Description	Recommended Action
High	4 undercut faces	Undercuts along draw direction [0,0,1] will lock the part in the mould without side-action tooling.	Add mechanical side-actions/lifters or split the tool along a secondary parting line; re-orient draw if possible.
Medium	2 zero-draft faces	Faces with no draft will scuff, gall, or resist ejection, increasing cycle time and part damage risk.	Add minimum 1–2° draft to all vertical faces along the pull direction.
Medium	Non-uniform wall ($\bar{A} = 5.30\text{ mm}$ across 2.53–3.54mm)	Wall thickness transitions can create sink marks, differential shrinkage, and warpage on this large flat/curved panel.	Smooth wall transitions with generous ribs/radii and target "d 10% thickness between adjacent sections.
Medium	Large projected area single-cavity tool (10,946.8 cm²)	Very large flow length from a limited gate count risks weld lines, hesitation marks, and uneven pack pressure across the panel.	Use multiple sequential valve-gated hot-drops with flow simulation to balance fill and hide weld lines in low-visibility zones.
Low	Threaded feature	Thread feature likely for a mounting insert/clip boss; direct-threaded plastic bosses can strip under vibration load.	Use moulded-in brass threaded inserts or self-tapping boss design rather than direct plastic threads.

§6 — DFM Issues (injection_moulding)

Severity	Area	Description	Impact	Fix
High	Undercut faces (x4)	Undercut geometry along the primary draw direction blocks straight-pull ejection.	Requires side-actions/lifters, increasing mould cost by 15–25% and cycle time by 2–4s per shot.	Redesign undercut regions or add mechanical core-pulls/lifters synchronized to the ejection stroke.
Medium	Zero-draft vertical walls	2 faces have no draft angle, causing ejection drag marks and potential part sticking.	Increased scrap rate and mould maintenance (polishing/textured wear).	Apply minimum 1.5° draft on all faces parallel to pull direction.
Medium	Non-uniform wall section (2.53–3.54mm)	Local thickness variation up to ~40% promotes differential cooling shrinkage.	Visible sink marks and warpage on the large styled surface, risking cosmetic rejects.	Core out thick sections to maintain nominal 2.5–2.8mm wall and add ribbing instead of solid mass.

Severity	Area	Description	Impact	Fix
Medium	Large single-cavity flow length	Projected area of ~1.1m² fed from limited gates on a thin 2.5mm wall risks hesitation and weld lines.	Visible weld lines on Class-A painted surface, potential short-shots at high volume cycle rates.	Use sequential valve-gated hot runner with 4–6 drops positioned per mould-flow simulation.
Low	Threaded boss feature	Direct-moulded thread in PP is prone to stripping under repeated fastening.	Field failures/warranty claims on bracket attachment points.	Specify moulded-in brass insert or ultrasonic-welded insert boss.

\$7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£6.20	£9.50	£15.00

Net Weight	2.163 kg	Material	mat-pp
Recommended Process	injection_moulding	Cycle Time	0.0128 hr/part
Setup Time	0.750 hr	Operations	0 ops

Operations Detail

Process-Specific Parameters

Casting Subtype	sand
Die/Mould Cost	£0
Die Life	0 shots
Cavities	0
Yield	0.0%
Flash Weight	0.000 kg
Yield	0.0%
Die Cost	£0
Strokes	0
Cavities	1
Wall Thickness	2.5 mm
Mould Cost	£200,000
Mould Life	500,000 shots
Projected Area	10946.8 cm²

§8 — AI Analysis Explanation

BUMPER.stp is a large (1.69m x 0.65m x 0.53m) thin-shell (2.5mm nominal wall) free-form panel (99% B-spline surfaces) consistent with an automotive bumper fascia. Resolving the UNDECIDED injection-moulding inputs as 'Moulded plastic': material selected as unfilled/impact-modified PP (TPO), single-cavity tool sized to the ~1.1m² projected area, and cycle time estimated from wall-thickness cooling. Key manufacturability risks are the 4 undercuts and 2 zero-draft faces requiring side-actions/lifters, non-uniform wall thickness risking sink and warpage, and the large flow length across a single cavity risking weld lines on the Class-A surface.

Stage-1 pre-selection: injection_moulding (85%) · sheet_metal · thermoforming